

AREFIN AZAM

☎ 703-474-3583 ✉ arefinazam20@gmail.com

Education

Stony Brook University

Bachelor of Eng./Masters of Sci. in Computer Engineering

Jul 2025 (B.E.) – May 2026 (M.S.)

GPA: 3.3 (B.E.), 3.5 (M.S.)

Relevant Coursework

Advanced Digital Logic Design, Digital Design Using VHDL, Computer Architecture, Parallel Processing Architectures, Embedded Microcontroller Systems, VLSI Design, Real-time Operating Systems, Semiconductor Physics, Deep Learning

Professional Experience

Tekno Visual

Junior Project Manager / Developer

Jan 2024 – May 2025

Fairfax, VA

- * Coordinated cross-functional technical work by tracking task dependencies, timelines, and deliverables across a 5+ member team.
- * Conducted regular progress reviews, identified bottlenecks, and helped resolve execution issues to maintain on-time delivery.
- * Prepared and presented structured technical summaries and status updates to senior stakeholders.

Projects

Multi-Digit Seven-Segment Display Driver with Write-Only SPI

Spring 2024

- * Designed a real-time, multiplexed seven-segment display controller in VHDL supporting multi-digit numeric output.
- * Implemented a write-only SPI interface with precise timing and control sequencing.
- * Built a self-checking testbench and a parameterized RTL architecture for scalable digit counts.

Matrix-Matrix Multiplier Hardware Accelerator

Fall 2024

- * Designed and implemented a parameterized hardware accelerator for matrix-matrix multiplication (MMM) using SystemVerilog.
- * Architected a modular dataflow design consisting of a matrix input controller, a pipelined multiply-accumulate (MAC) datapath, and a FIFO-based output buffer for product storage.
- * Integrated AXI-Stream interfaces to enable high-throughput, streaming input/output of matrix data.
- * Optimized system throughput via pipeline scheduling, control logic refinement, and back-pressure management.

Four-Stage Pipelined Multimedia Unit

Fall 2024

- * Designed and implemented a four-stage pipelined multimedia ALU in SystemVerilog.
- * Implemented data hazard detection and forwarding logic to handle read-after-write dependencies and improve pipeline throughput.
- * Developed a custom assembler to translate 25 custom MIPS-style instructions into machine code compatible with the pipeline datapath and control logic.
- * Verified functional correctness using a self-checking testbench and waveform analysis.

GPU-Accelerated Single-Source Shortest Path

Fall 2025

- * Implemented a CUDA-based tiled matrix multiplication kernel, achieving up to 35,000× speedup over a sequential CPU baseline.
- * Designed a GPU-resident wavefront shortest-path algorithm using parallel frontier relaxation.
- * Resolved parallel data hazards using atomic operations and double-buffered distance arrays.
- * Optimized global-memory access patterns for memory coalescing and throughput.
- * Visualized shortest paths using OpenCV and validated results against a CPU baseline.

Pipelined 8-Bit Carry Select Adder (Custom CMOS Design & Layout)

Fall 2025

- * Designed a pipelined synchronous 8-bit carry select adder targeting 4 GHz operation, optimizing the carry critical path using a two-block (4+4) CSA architecture.
- * Completed full custom layout in Cadence Virtuoso, achieving DRC and LVS clean verification and accurate post-layout simulations.
- * Verified functional correctness through transient and decimal-converted simulations and evaluated average power consumption under realistic switching activity.

Skills

Languages: Python, C, C++, Java, MIPS Assembly, VHDL, SystemVerilog, Matlab, CUDA, OpenMP, ISPC

Developer Skills: OrCAD, Microchip Studio, Active-HDL, Synplify, Questasim, LVS, DFS, Cadence Virtuoso, Altium Designer, Operating Systems, Windows, Linux, Command Line Interface, USART, I2C, SPI, 2D/3D Design